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21 AUGUST 1975
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21 October 1968

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, MAGNETRON

TYPE 6551

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Fixed frequency, 24,035 ±235 MHz, pulse type, integral magnet, rated peak power output 40 kw.

ABSOLUTE RATINGS:

Parameter:	Ef	If(surge)	tk	epy	ib	tpc	Du	pi	T (anode)
Unit:	V	a	sec	kv	a	μs	----	kw	°C
Maximum:	6	12	---	16	18	0.5	0.0007	270	130
Minimum:	--	---	180	--	10	0.1	----	---	--- (see note f)

PHYSICAL CHARACTERISTICS:

Dimensions:	See figure 1	Output coupling:	See figure 1
Mounting position:	Any (see note 4)	Cathode:	Unipotential
Mounting support:	See figure 1	Weight:	7.5 pounds (approx)
Input coupling:	See figure 1		

TEST CONDITIONS:

Parameter:	Ef	tk	tpc	Du	trv	ib	VSWR
Unit:	V	sec	μs	----	μs	a	----
Tolerance:	±10%	---	±10%	----	±0.01	--	----
Test 1:	5	180(max)	---	----	---	--	----
Test 2:	0 (see note 3)	---	0.15 (see note 8)	0.0006	0.09	15	1.1:1 (max)

REQUIREMENT OR TEST

GENERAL

First article inspection - Required (Appendix F)

6551

SC 5960

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METHOD	REQUIREMENT OR TEST	CONDITIONS	SYMBOL	LIMITS MINIMUM	LIMITS MAXIMUM	UNIT
	Qualification					
4028	Barometric pressure, reduced	Pressure = 10 psia (min)	---	---	---	---
1031	High-frequency vibration	No voltages; 5G; F = 10 to 150 Hz; t = 300 seconds; See note 6	---	---	---	---
1042	Shock	Condition A	---	---	---	---
----	Fixed tuned frequency	T (anode) = 120 ±5°C	F	23,750	24,225	MHz
4027	Temperature coefficient	T (anode) = 60 to 100°C	ΔF/ΔT	---	0.6	MHz/°C
1047	Low-temperature performance	See note 7	---	---	---	---
1047	High-temperature performance	See note 7	---	---	---	---
	Quality conformance inspection, part 1					
1301	Heater current	Test Condition 1	If	2.6	3.2	A
4003	Pressurization	Pressure = 40 psia	---	---	---	---
430c	Pulse voltage		epy	13	15	kv
4250	Power output		Po	24	---	W
4308	RF bandwidth	See note 9	BW	---	2.25 tpc	MHz
----	Fixed tuned frequency	T (anode) = 40 o 60°C	F	23,800	24,270	MHz
	Quality conformance inspection, part 2					
4310	Frequency pulling		ΔF	---	30	MHz
4315	Stability	BW = 100 MHz; VSWR = 1.5:1; See note 10	MP	---	1	%
1031	Low-frequency vibration	No voltages	---	---	---	---
	Quality conformance inspection, part 3					
----	Life-test provisions	Group D; VSWR = 1.5:1; See notes 4, 5 and 11	---	350	---	cycles
----	Life-test end points:					
4250	Power output		Po	19.2	---	W
4306	Pulse voltage		epy	---	16	kv
4308	RF bandwidth		BW	---	2.7 tpc	MHz
4315	Stability		MP	---	2	%

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NOTES:

1. The AQL for all tests listed under quality conformance inspection, part 1, shall be 1 percent, inspection level of II.

2. The AQL for all tests listed under quality conformance inspection, part 2, shall be 6.5 percent, inspection level of S3.

3. The heater voltage shall be reduced to zero during operation after the application of high voltage.

4. The tube shall be mounted with the cathode vertical during all tests except life test. The tube shall be mounted in a horizontal position during life test.

5. The tube shall be cycled in accordance with the following schedule:

Test condition	Duration
1	3 minutes
2	60 minutes
Off	15 minutes

6. Vibration shall be along the X, Y, and Z axes for periods of 5 minutes. In each plane, the frequency vibration shall vary from minimum to the maximum and returned to minimum in the time specified.

7. With the exception of holding period, the only criterion for this test is that the tube shall exhibit normal operation 30 seconds after snap-on.

8. No spike or ripple shall exceed ±7 percent of the average peak value of voltage or current. Inverse voltage shall not exceed 20 percent of the forward voltage. The negative amplitude of any post-pulse voltage oscillations shall not exceed 10 percent of the average peak value.

9. Stability shall not be measured under this test. The RF bandwidth shall be within the limits specified when a VSWR of 1.5:1 is introduced in the load at a distance of approximately 0.5 meters from the magnetron coupling flange, the phase being adjusted for operation in the sink.

10. Satisfactory operation shall be obtained for the last 2 minutes of a test interval not to exceed 5 minutes. Observations shall be made for load phase positions corresponding to maximum power and to minimum power.

11. The standing wave introducer shall be moved during the test so that operation is obtained for load phase positions corresponding to maximum power output, minimum power output, maximum frequency, and minimum frequency. If automatically driven, the standing wave introducer shall be cycled continuously through a line length approximately one-half wave length long at a maximum rate of four cycles per hour; if manually moved, the standing wave introducer shall be cycled through the four load phase positions corresponding to maximum power, minimum power, maximum frequency, and minimum frequency at least once during the specified life of the tube, spending approximately equal periods of time in each phase position.

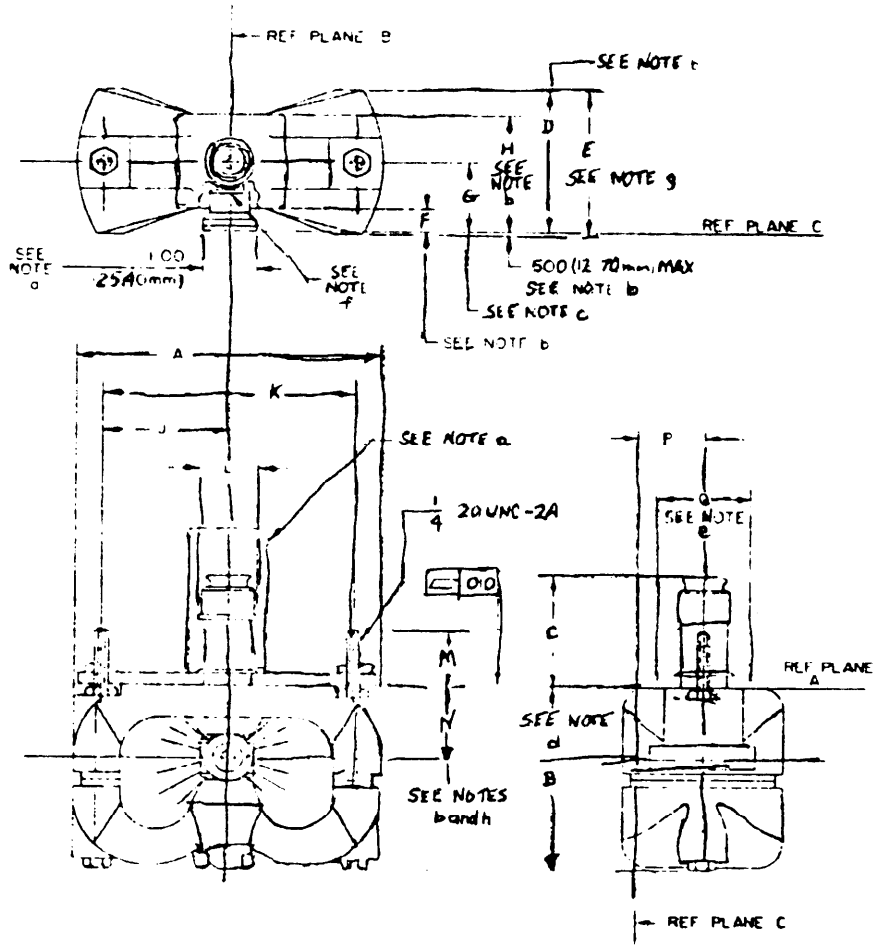
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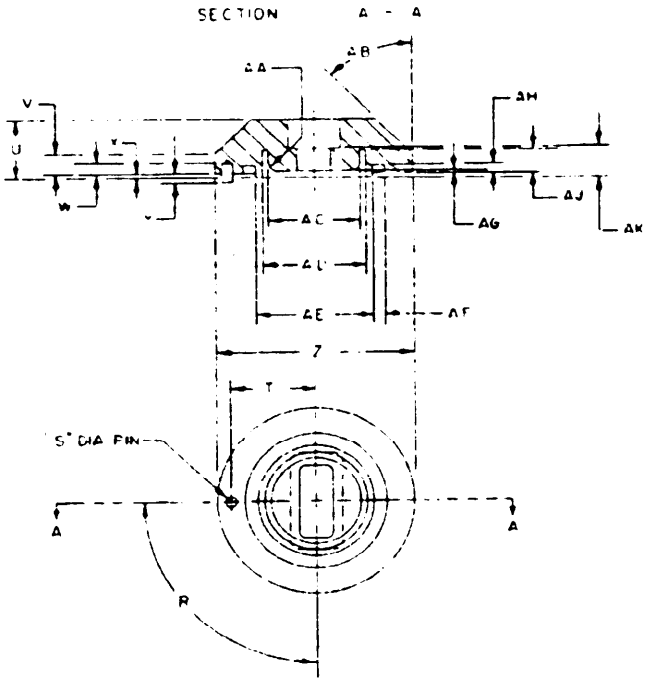
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NOTES:
a. Guard used for shipping purposes only.
b. Includes angular as well as lateral deviations.
c. This dimension applies to location of mounting bolts with respect to reference plane C.
d. This dimension applies either to head of magnet bolt, top of exhaust cap or magnet coating.
e. This dimension applies to the flat on top of magnet.
f. Temperature is taken at this point.
g. This refers to the size of coated magnet only.
h. Centering of drilled holes in output flange may vary from true location by .007 (.18 mm).

FIGURE 1. Outline Drawing For Type 6551

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses		Notes
	Minimum	Maximum	
Quality conformance inspection, part 1			
A		5.875 (149.23)	
B		3.750 (95.25)	d
C		2.125 (53.98)	
D		3.000 (76.20)	b
E		3.187 (80.95)	g
K	4.640 (117.86)	4.740 (120.40)	
L		1.165 (29.59)	
R	89° 45'	90° 15'	
S	.061 (1.55)	.062 (1.57)	
T	.429 (10.90)	.432 (10.97)	

Ltr	Dimensions in inches with metric equivalents (mm) in parentheses		Notes
	Minimum	Maximum	
Quality conformance inspection, part 2			
F	.281 (7.14)		b
G	1.205 (30.61)	1.265 (32.13)	c
H		2.312 (58.72)	b
J	2.315 (58.80)	2.375 (60.33)	
M	.470 (11.94)	.656 (16.66)	
N	1.413 (35.89)	1.453 (36.91)	b,h
P	1.110 (28.19)	1.360 (34.54)	
Q	1.710 (43.43)	1.840 (46.74)	e
U	.310 (7.87)	.320 (8.13)	
V	.120 (3.05)	.130 (3.30)	
W	.060 (1.52)	.070 (1.78)	
X	.020 (.51)	.025 (.64)	
Y	.050 (1.27)	.055 (1.40)	
Z	.995 (25.27)	1.005 (25.53)	
AA	.006 (.15)	.010 (.25)	
AB	44° 30'	45° 30'	
AC	.470 (11.94)	.474 (12.04)	
AD	.534 (13.56)	.538 (13.67)	
AE	.603 (15.32)	.605 (15.37)	
AF	.060 (1.52)	.064 (1.63)	
AG	.005 (.13)	.006 (.15)	
AH	.050 (1.27)	.054 (1.37)	
AJ	.127 (3.23)	.131 (3.33)	
AK	.150 (3.81)	.160 (4.06)	

FIGURE 1. Outline Drawing For Type 6551 - Continued

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